MATHS I · WEEK 1

Relations: the 4 properties and their traps

Reflexive, symmetric, antisymmetric, transitive. Asked in almost every paper.

DEFINITIONS

R is a relation on A

Reflexive

$(a, a) \in R$ for every a in A

Symmetric

$(a, b) \in R \Rightarrow (b, a) \in R$

Antisymmetric

(a, b) and (b, a) in $R \Rightarrow a = b$

Transitive

(a, b) and (b, c) in $R \Rightarrow (a, c) \in R$

Equivalence

reflexive + symmetric + transitive

TRAPS

Where marks are lost

⚠ If (1, 2) and (2, 1) are in R, transitive needs (1, 1) and (2, 2) too.

⚠ Reflexive needs (a, a) for every element, even ones that never appear in R.

⚠ Symmetric and antisymmetric are not opposites. A relation can be both, or neither.

⚠ Tip: write R as a list of pairs first. Then test each property.

PYQ · MAY 2023

Your turn

A × A has 9 elements. Two of them are (2, 0) and (0, 8). What is the sum of all the elements of A?

Comment your answer, then swipe.

ANSWER

10

1. $|A \times A| = |A|^2 = 9$, so A has 3 elements.
2. (2, 0) and (0, 8) put 2, 0 and 8 in A. So $A = \{0, 2, 8\}$.
3. Sum = 0 + 2 + 8 = 10.

Practise real Qualifier PYQs free.

Timed mocks on the real exam screen, graded with the official answer keys.

btechi.in →

Save this post for revision. Share it with a friend writing the Qualifier.

Not affiliated with IIT Madras.